Biophysical Dynamics & Autonomous Operator Discovery
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October 7, 2026

The Editors • *Nature Communications*
Springer Nature, 4 Crinan Street, London, N1 9XW, United Kingdom

SUBJECT: Submission of Original Research Article — "General Scaling Laws and Thermodynamically Consistent Reduced Stochastic Operators for Macromolecular Protein Dynamics"

Dear Editors,

We are pleased to submit our original research manuscript entitled **"General Scaling Laws and Thermodynamically Consistent Reduced Stochastic Operators for Macromolecular Protein Dynamics"** for consideration as an Article in *Nature Communications*.

A central challenge in computational biophysics and statistical mechanics is constructing low-dimensional equations of motion from atomistic molecular dynamics (MD) trajectories that retain true physical kinetics. Standard coarse-graining approaches frequently encounter two fundamental compromises: discrete Markov state models depend sensitively on microstate discretization and lag times, whereas continuous Brownian or unconstrained neural surrogates routinely violate microscopic detailed balance and neglect non-Markovian memory friction near transition barriers.

In this work, we present a continuous, structure-preserving stochastic generator framework evaluated across seven benchmark systems spanning model peptides (dialanine, trialanine) to five structurally diverse proteins (d = 2 to 150 conformational dimensions: Chignolin, Trp-cage, Villin HP35, GB1, and Human Ubiquitin). Our primary findings include:

1. **Sub-linear operator rank scaling:** The effective dynamical rank of the diffusion operator scales sub-linearly with chain length ($r_{90}(d) = 0.745 \cdot d^{0.887}$, with a 95% non-parametric bootstrap confidence interval of [0.830, 0.950], rejecting linear growth at $P(\alpha < 1.0) = 99.95\%$). In contrast, an uncoupled harmonic chain null model recovers linear scaling ($\alpha_{null} \approx 1.01 \pm 0.02$), demonstrating that dynamical compression arises from cooperative macromolecular interactions.
2. **Dynamic locality and tertiary contact gating:** Off-diagonal diffusion couplings decay as a power law along the sequence ($|i - j|^{-1.34}$), but native tertiary contacts gate significant off-diagonal coupling with a 2.1- to 3.9-fold enrichment over non-contact pairs.
3. **Asymptotic linear-time estimation:** Exploiting this contact-gated structure yields an $O(d^{0.99})$ sparse estimator that accelerates generator estimation by 28-fold at d = 2,400 with under 1.8% relative spectral error.
4. **Invariant interior mobility floor:** Core residues converge to an invariant mobility plateau ($\bar{D}$ = 0.0496 ± 0.0016 rad^2/ps, CV 8.0%) across all five protein folds. Computed dihedral mobilities correlate significantly with experimental 1.8 Å X-ray crystallographic Debye-Waller B-factors (r_s = 0.272, p = 0.019).
5. **Thermodynamic consistency:** Incorporating structure-preserving GENERIC drift eliminates 90.8% of artificial steady-state entropy production present in unconstrained estimators, while extended Mori-Zwanzig state space embedding (z, v, w) reproduces atomistic folding/unfolding barrier-crossing times within the 95% bootstrap confidence interval.

Because this study addresses foundational questions at the interface of non-equilibrium statistical mechanics, macromolecular kinetics, and data-driven physical modeling, we believe it will interest the broad readership of *Nature Communications*, particularly researchers in computational biophysics, structural biology, and applied mathematics.

The manuscript is supported by a comprehensive Supporting Information Appendix detailing mathematical proofs, numerical benchmark matrices, and algorithmic implementations. All simulation trajectories, analysis pipelines, and software will be made fully accessible under an open-source license upon publication. This work is original, has not been published previously, and is not currently under consideration elsewhere. Both authors have approved the submission.

Thank you for your time and editorial evaluation.

Sincerely,

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